

PROJECT		JOB NO.				
CONTRACTOR :						
SUB-CONTRACTOR : Winner Engineering Pte Ltd						
TRADE	: ACMV SERVICES		DRAWING NO :			
FUNCTIONAL- TESTING COMMISSIONING CHECKLIST (COOLING TOWER)						
Unit Designation :						
Unit Make :						
Model no.						
Location :						
Serial no :						
S/N	Section	Inspection / Test Item	CHECKED CONDITION (Please tick inside checkbox)			Remarks
			Yes	No	NA	
1	General	Check that the equipment is clean, free of debris, and undamaged before commencing operations.				
2	Documents	Verify chiller tag, make, model, serial number, refrigerant and rated capacity.				
3	Electrical	Verify supply voltage, phase sequence and running current on applicable phases.				
4	Controls	Test local start/stop, Auto/Manual, Remote/BMS and run/trip indications where provided.				
5	BMS	Compare BMS commands, statuses and analog readings with actual field conditions.				
6	Fill & leakage	Fill basin and circulate water; inspect basin and pipe connections.				
7	Makeup	Test automatic makeup-water valve through the operating level range.				
8	Level alarms	Test low/high-level alarms and associated pump/fan interlocks by approved simulation.				
9	Drainage	Verify overflow and drain flow paths by approved water test.				
10	Treatment	Test conductivity controller, blowdown valve and dosing interlocks where installed.				
11	Distribution	Run water circulation and verify spray/distribution over fill.				
12	Water flow	Measure condenser-water flow to each operating tower cell and balance as required.				
13	Fan	Verify fan rotation and airflow direction using manufacturer-approved start procedure.				
14	Drive	Test each fitted fan speed or VSD control range.				
15	Mechanical	Check noise, vibration, bearings and gearbox temperatures where applicable.				
16	Vibration trip	Test vibration switch alarm/trip where fitted by approved method.				
17	Thermal control	Verify fan staging/modulation against condenser-water temperature demand.				
18	Thermal duty	Record entering/leaving water temperature, water flow and entering-air wet-bulb temperature.				
19	Thermal duty	Check tower range and approach from the recorded temperatures where specified.				
20	Performance	Verify specified thermal performance at agreed test conditions.				
21	Carryover	Inspect drift eliminators, louvers and surrounding area during operation.				
22	Plant sequence	Test cell staging, isolation valves, duty/standby and chiller/CW-pump coordination where provided.				

23	Recovery	Verify fan fault, shutdown and power-restoration responses as specified.				
24	Stability	Complete specified running period and treatment-system checks.				
25	Completion	Record final setpoints, control mode and operating settings; restore normal operation.				

Comments to follow up Action :

Yes = satisfactory; No = rectify and re-inspect; NA = not applicable (state reason). Blank = not checked.

CHECKED BY:		WITNESSED BY	INSPECTED BY
NAME			
DESIGNATION :			
COMPANY :			
SIGNATURE :			
DATE :			